

Stitiprangya Patra

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EXPERIENCE

Member of Technical Staff - 1 (MTS-1) 07/2025 – Present
Nielsen Media India

- Architected distributed, fault-tolerant backend workflows using **Python** and **Java**, processing nearly **1.3 TB** daily across **15+** datasets; designed modular, configurable execution pipelines to ensure deterministic behavior, safe reprocessing, and scalable downstream service integration.
- Re-architected legacy rollup and monthly processing workflows by redesigning core business logic using **Python**, **Java**, and **Amazon Redshift SQL**, introducing dependency-aware orchestration and retry-safe execution to ensure idempotent partition processing; reduced **AWS** compute costs by **30%** while improving end-to-end latency by **10%**.
- Engineered aggregation pipelines for **rollup and monthly** batch processing, optimizing validation logic and ensuring correctness of partitioned data loading into downstream data models (MDL).
- Improved system reliability by resolving schema drift, malformed payloads, and null edge cases; implemented metadata validation and deterministic rolling-window logic, improving SLA adherence from **85%** to **95%** and reducing failure rates by **35%**.

Software Engineer Intern 01/2025 – 06/2025
Rahi Platform Technologies

- Built backend services for a financial SaaS platform (Loan Management System, LOS, Handoff Module) using **Java 21** and **Spring Boot**, following hexagonal architecture to enable modular, testable service design.
- Architected transaction-safe penal charge calculation engine using **Java 21**, implementing configurable multi-state tax compliance models (IGST/CGST/SGST) with concurrency-safe processing of **5000+** daily financial transactions, achieving **99.9%** monetary accuracy and reducing compliance issues by **30%**.
- Optimized REST APIs and CRUD operations with Java and Spring Boot, reducing average response time from **420ms to 280ms (33% improvement)** and doubling system throughput through database query optimization and connection pooling.
- Designed and built 21 REST API endpoints for loan account management with Spring Boot and Spring Data JPA, achieving **95%** test coverage using **JUnit** and **Mockito**, ensuring production-ready code quality.

AI/ML Intern 07/2024 – 12/2024
TIF Labs

- Developed **3-layer** CNN model for damaged car part detection achieving **85%** accuracy on **10K+** images; reduced overfitting by **15%** via data augmentation, dropout, and hyperparameter tuning; integrated **LangChain** with **AWS Bedrock** (Claude) for automated damage severity assessment.
- Built web scraping pipeline using **BeautifulSoup** to extract and compare competitor data across **20+** websites, reducing manual analysis time by **80%**; leveraged LLMs for intelligent data comparison and AI-driven image quality evaluation.
- Implemented real-time damage detection using **OpenCV** and **YOLOv8** achieving **92%** mAP; developed **Flet**-based segmentation application and integrated REST APIs for automated customer notifications, enabling production-ready deployment.

Software Engineer Intern 05/2023 – 06/2023
Perceptiviti Data Solutions

- Enhanced Sherlock auto-adjudication system for medical insurance fraud detection, resulting in **5.4 million AED** saved for major insurer; improved ML/DL models using **DNN**, **Random Forest**, and **XGBoost**, achieving **25%** improvement in fraud detection accuracy and 12% reduction in false positives.
- Automated ETL pipeline for Oracle-to-MySQL data migration using Python, processing 100K+ daily records and reducing manual errors by **10%**; developed 15+ business validation rules using Python and SQL to verify claims against medical necessity guidelines, improving system efficiency by **20%**.

EDUCATION

Birla Institute of Technology and Science (BITS Pilani), Goa Campus — B.E. (Electronics & Communication) + M.Sc. Mathematics (2020–2025) | CGPA: **7.12/10.0**
Delhi Public School Bangalore East — CBSE Class XII: **95.4%** | Class X: **95.2%**

PROJECTS

- ArthaMitra – AI Financial Digital Twin Platform (Python, Streamlit, FastAPI, Plotly, Pandas):** Built an AI-powered financial digital twin platform that models financial health, simulates life events, and generates personalized wealth recommendations through real-time scenario analysis using modular financial, behavioral, prediction, and recommendation engines.
- VoxBridge – Offline AI Video Translation Platform (Python, FastAPI, Streamlit, Faster-Whisper, NLLB-200, FFmpeg):** Built an offline multilingual AI translation platform supporting video, audio, and text translation with subtitle generation, AI voice synthesis, and media processing for files up to **2 GB**.

TECHNICAL SKILLS

Programming Languages: Java, Python, C++, SQL
Backend & Frameworks: Spring Boot, FastAPI, REST APIs, Microservices, Hexagonal Architecture, Solid Principles
Core CS: Data Structures & Algorithms, Object-Oriented Design, Low-Level Design, Distributed Systems Fundamentals
Data Engineering: Apache Airflow, MySQL, PostgreSQL, Oracle, Data Modeling, NumPy, Pandas, Data Analysis, Logging
Cloud & DevOps: AWS (S3, EC2, Lambda, RDS, SQS, SNS), Git, GitHub, CI/CD
ML/AI: TensorFlow, Scikit-learn, Random Forest, XGBoost, DNN/CNN, LangChain, OpenCV, YOLOv8, TF-IDF, NLTK
Testing: JUnit, Mockito, TestNG, Unit Testing, Integration Testing

ACHIEVEMENTS

Certifications: Neural Networks and Deep Learning, Structuring Machine Learning Projects, Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization (deeplearning.ai - Andrew Ng)
Marathon & Fundraising: Completed Airtel Delhi Half Marathon 10K, raising ₹52,500 for Tamanna Foundation
Athletics: BITS FC (Goa Division 1 & 3), Athletics Team, Chess Enthusiast